

ECEN 215, Spring 2022

Electrical Engineering Principles

Lecture 3: Kirchhoff's laws

Ramy Rady

Dept. of Electrical and Computer Engineering

Texas A&M University

ramyrady@tamu.edu



TEXAS A&M
UNIVERSITY

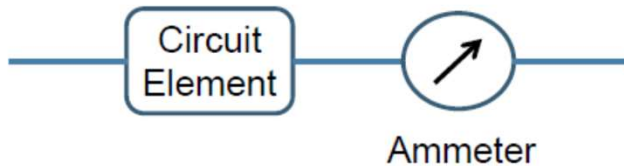
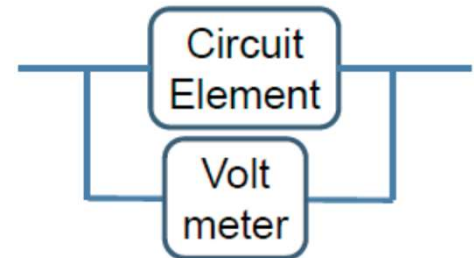
OUTLINE



- Electrical Sources
 - Series and Parallel
 - Kirchhoff's Laws
-

Measuring Voltage and Current

- Measure voltage with a voltmeter: probes *across* the two nodes you want to measure
- Measure current with an ammeter: break the circuit and place meter *in series* so that you can measure the current *through* it



Current is measured through an interface

Voltage is measured between two points

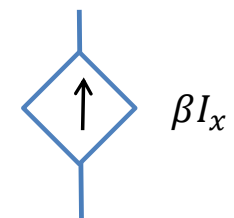
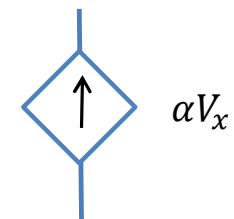
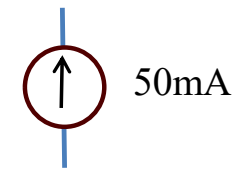
Electrical Sources

- **Definition:** An electrical source is a device capable of converting non-electric energy to electric energy.
- Battery – converts chemical to electrical energy
- Generator – converts mechanical energy to electrical energy
- Wind turbine – converts wind to electrical energy
- Solar cell – converts sunlight to electrical energy

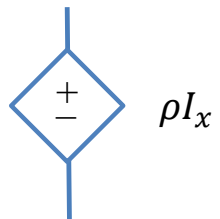
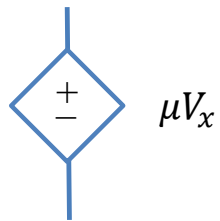
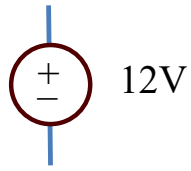


Current Sources

- **Definition:** A current source produces a constant current through its terminals regardless of the voltage across it.
- **Independent current source** – produces the same current regardless of any other voltage/current elsewhere in the circuit.
- **Voltage controlled current source** – produces a current which depends on V_x , some other voltage elsewhere in the circuit.
 - α = constant of proportionality in amps/volt
- **Current controlled current source** – produces a current which depends on I_x , some other current elsewhere in the circuit.
 - β = dimensionless constant of proportionality



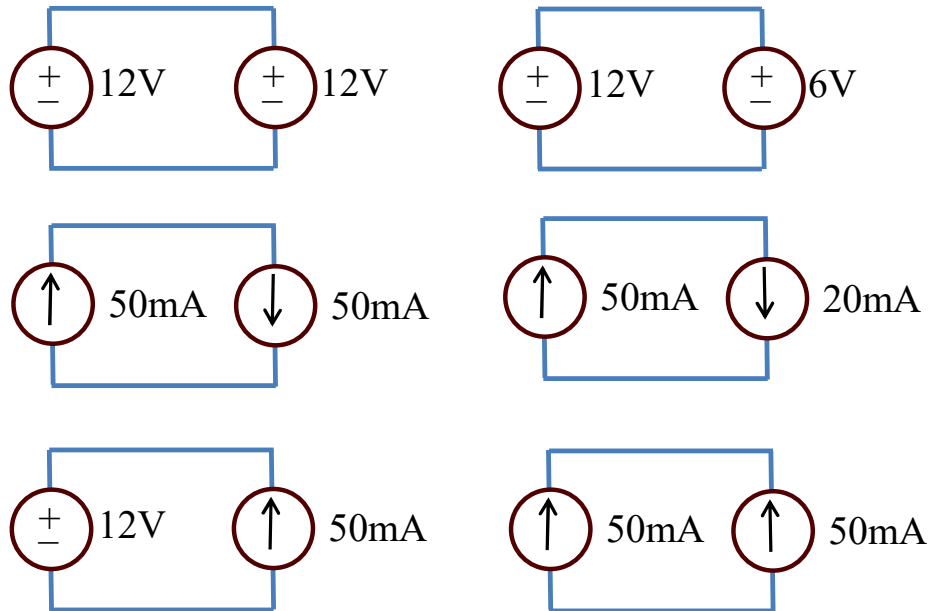
Voltage Sources



- **Definition:** A voltage source produces a constant voltage across its terminals regardless of the current flowing through it.
- **Independent voltage source** – produces the same voltage regardless of any other voltage/current elsewhere in the circuit.
- **Voltage controlled voltage source** – produces a voltage which depends on V_x , some other voltage elsewhere in the circuit.
 - μ = dimensionless constant of proportionality
- **Current controlled voltage source** – produces a voltage which depends on I_x , some other current elsewhere in the circuit.
 - ρ = constant of proportionality in volts/amp

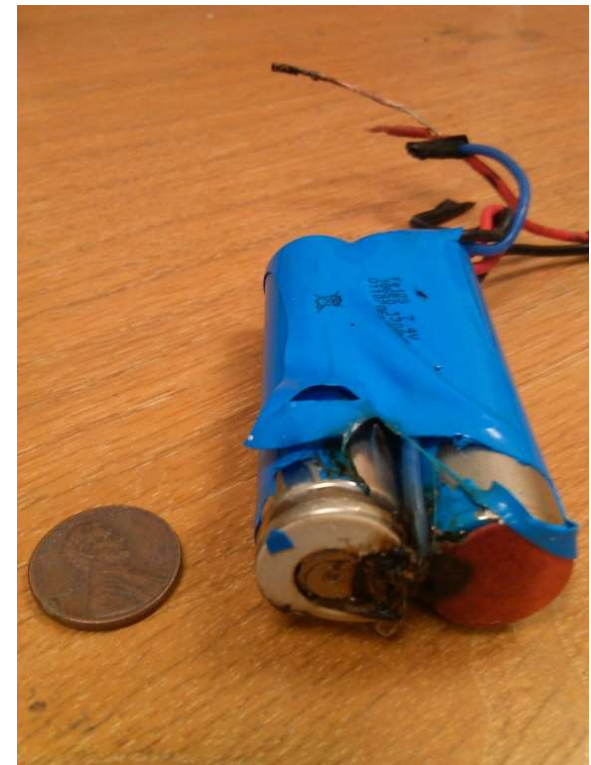
Restrictions on Source Connections

- Voltage and current sources cannot be connected in arbitrary manners. The following examples show both valid and invalid connections.



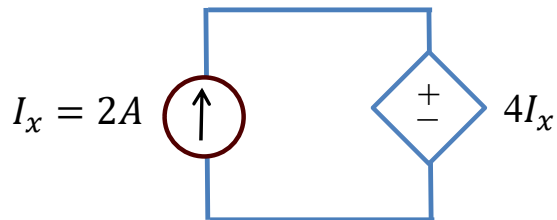
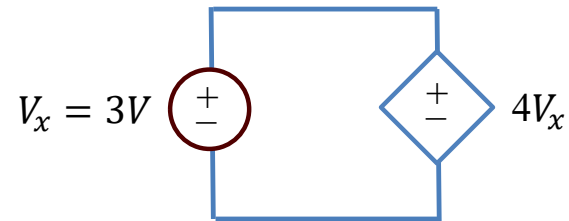
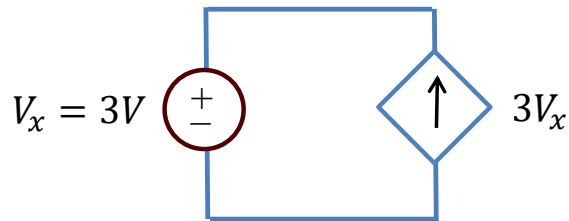
VALID

INVALID

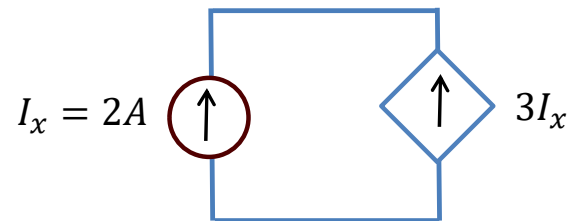


Restrictions on Source Connections

- Here are some more examples of both valid and invalid source connections.



VALID

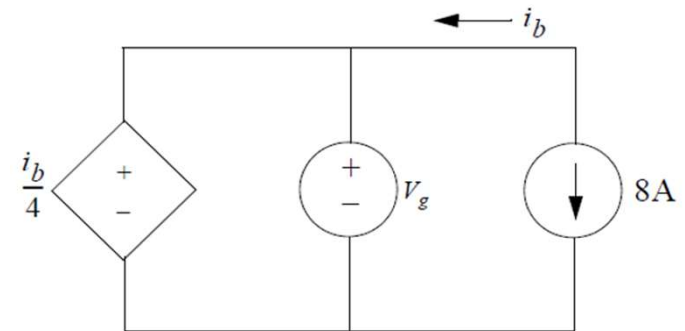


INVALID

Example of Restrictions on Source Connections



- For the circuit shown:
 - (a) What value of V_g is required in order for the interconnection to be valid?
 - (b) For the value of V_g found in part (a), what is the power associated with the 8A current source?

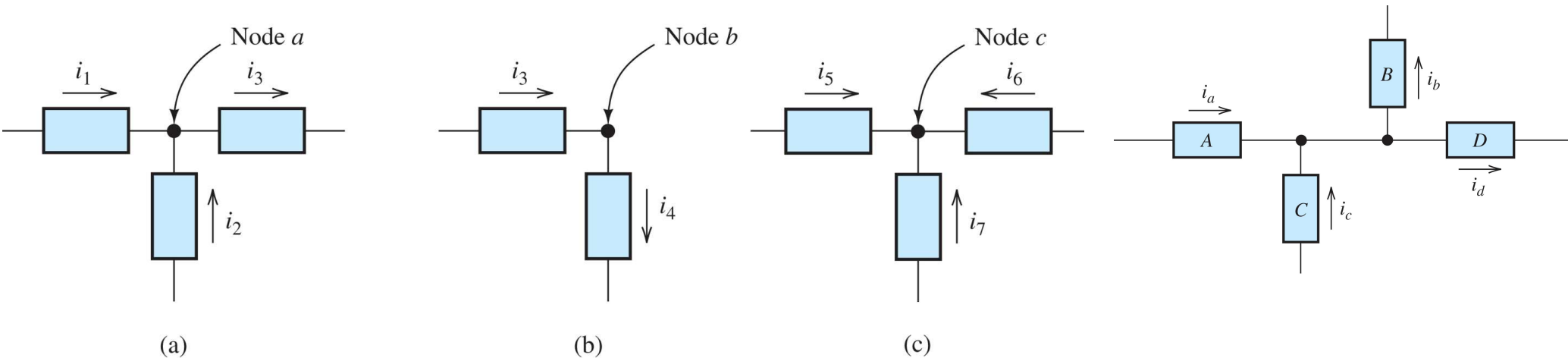


Node and Loop



Node: The connection point between two or more circuit elements is called a node.

Loop: Any closed path in the circuit is called a loop.



Node and Loop

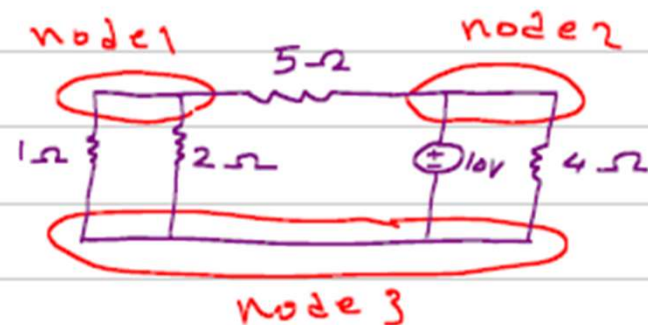


Ex. Identify the nodes, series elements and parallel elements.

3 nodes

$1\Omega \parallel 2\Omega$

$10V \parallel 4\Omega$

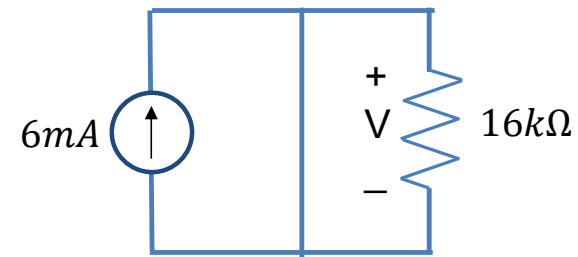


No series connected elements

Short Circuit and Open Circuit

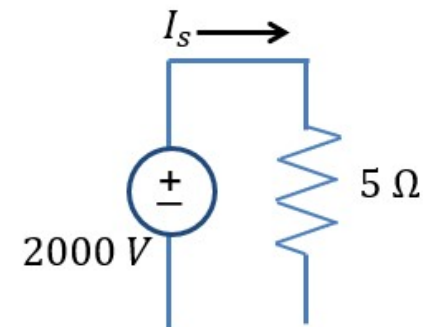
- The **voltage** across a short circuit is zero

- $V_{aa}=0$



- The **current** through an open circuit is zero

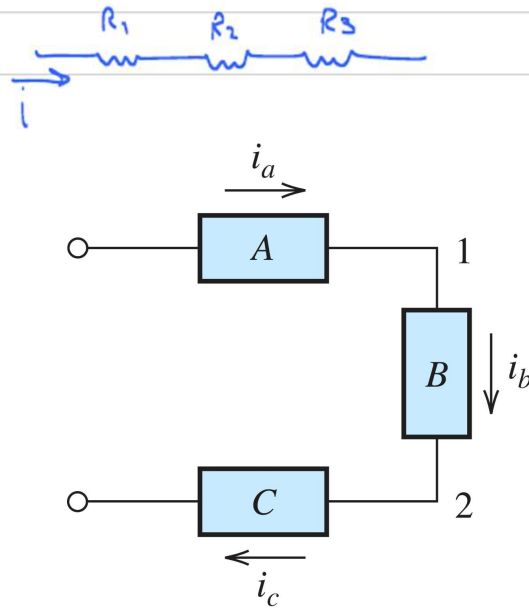
- I_s can't go through an open path



Series and Parallel



Series connected elements: When two or more elements are connected sequentially, they have the same current and are said to be in series.

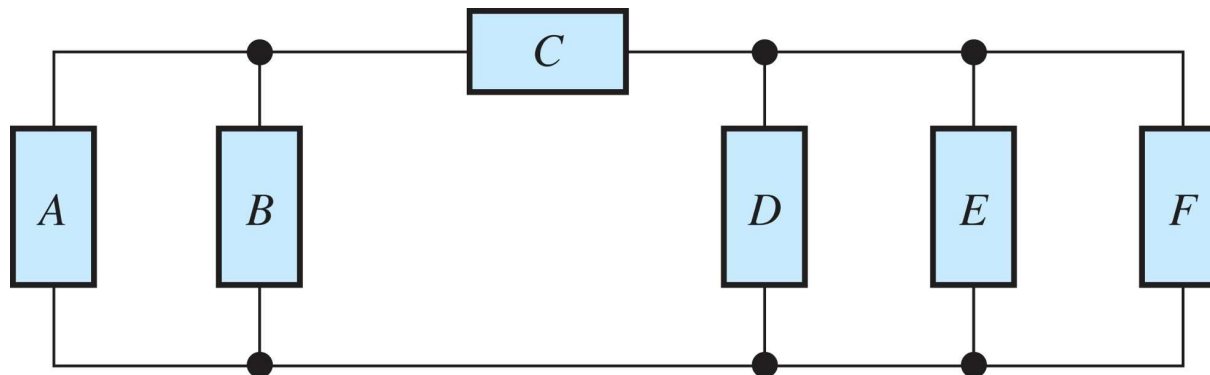
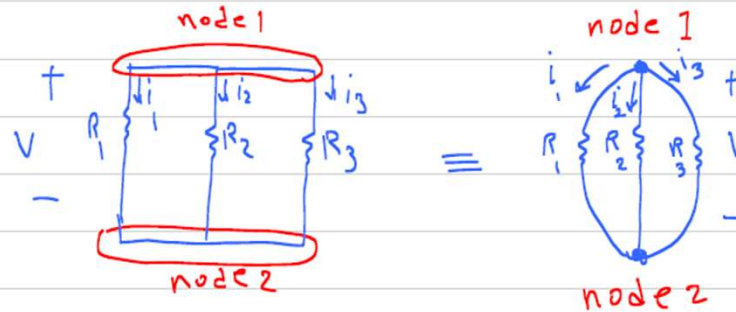


Copyright ©2018 Pearson Education, All Rights Reserved.

Series and Parallel

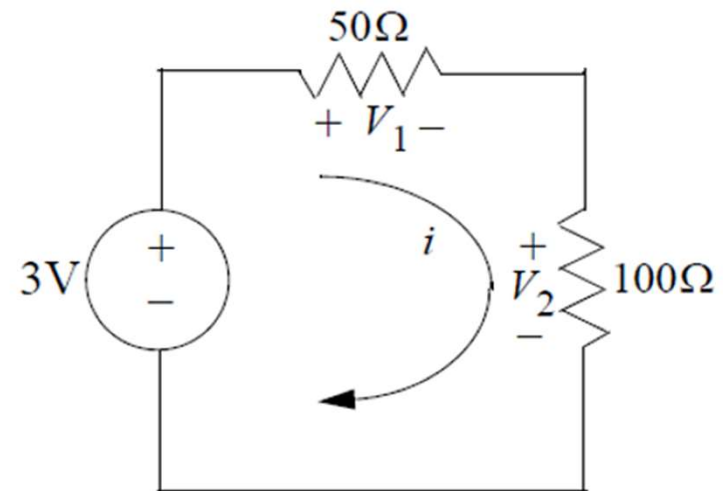


Parallel connected elements: When two or more elements are connected between the same node pair, they have the same voltage and are said to be in parallel.



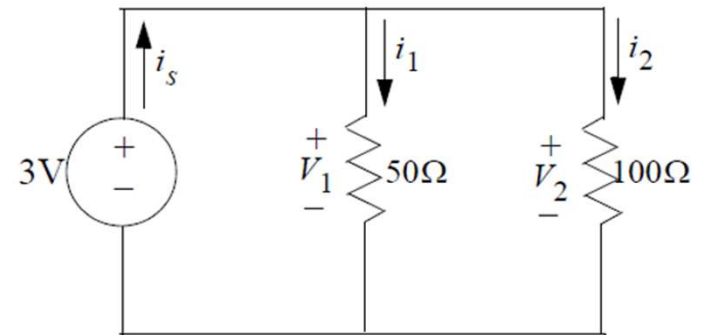
Example of Series Devices

- Elements in **series** share one junction, with nothing else connected to the same junction
- They have the **same current** (KCL)
- Resistors in series **add**
- For the circuit shown:
 1. Find the current flowing through the circuit.
 2. Find the voltage across each resistor.

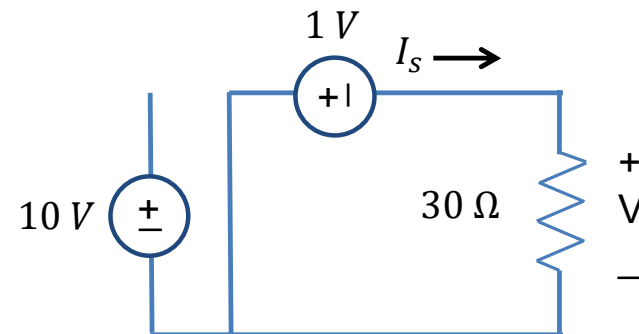
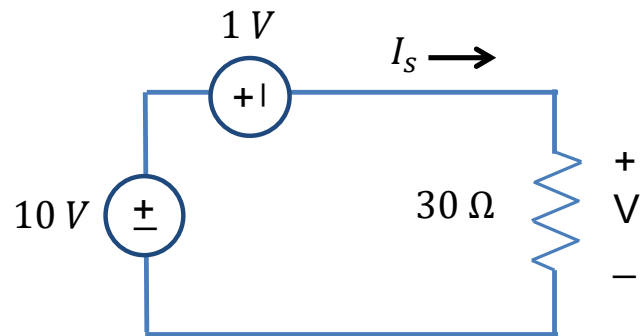
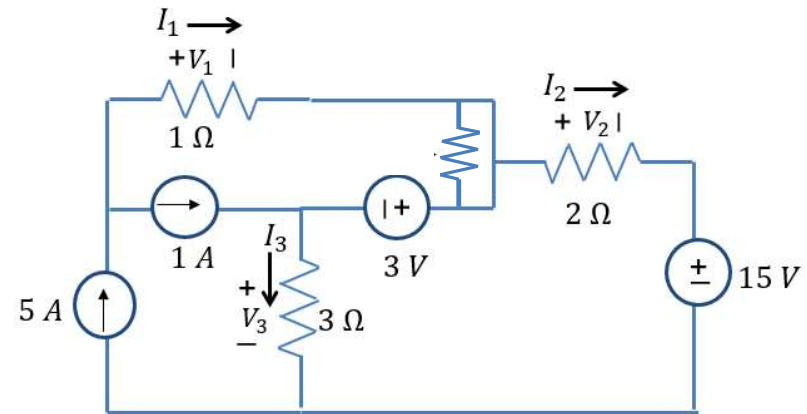
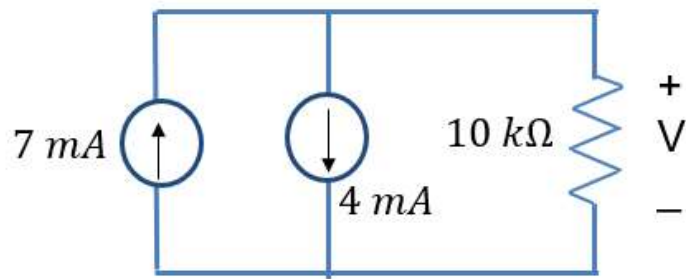


Example of Parallel Devices

- Elements in **parallel** share two junctions, regardless of what else is connected
- They have the **same voltage** (KVL)
- Resistors in **parallel** combine as: $R_{\parallel} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}}$
- For the circuit shown:
 1. Find the voltage drops across each resistor.
 2. Find the currents flowing through each resistor.
 3. Find the current flowing through the voltage source.

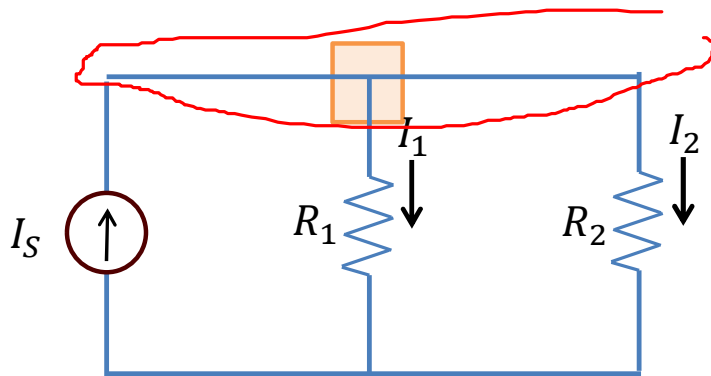


Examples: Find Shorts, Opens, Parallel, Series



Kirchhoff's Laws

Kirchhoff's Current Law (KCL): The sum of all currents entering (or leaving) any node of a circuit must be zero.



$$I_S - I_1 - I_2 = 0 \quad (\text{net flow in} = 0)$$

or equivalently

$$-I_S + I_1 + I_2 = 0 \quad (\text{net flow out} = 0)$$

or equivalently

$$I_S = I_1 + I_2 \quad (\text{flow in} = \text{flow out})$$

A Water Analogy: Consider water flowing through a series of pipes. At any junction where multiple pipes intersect, the net flow into the junction must be zero.

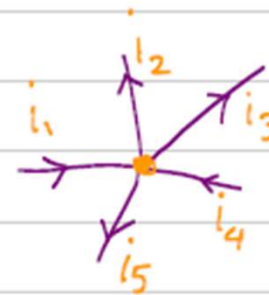


KCL



Kirchhoff's Current Law (KCL) : The algebraic sum of all currents entering a node (or a closed boundary) is zero.

$$\text{KCL:} \quad \sum_{n=1}^N i_n = 0$$



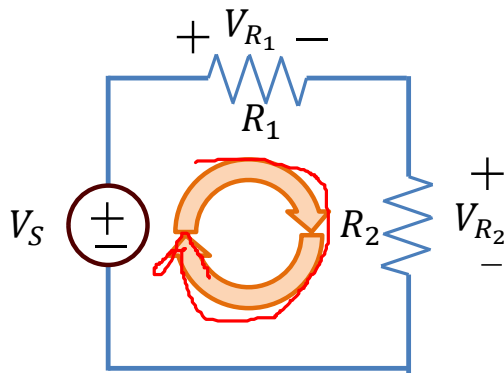
Taking the currents leaving the node positive.

$$\text{KCL:} \quad -i_1 + i_2 + i_3 - i_4 + i_5 = 0$$

Kirchhoff's Laws



Kirchhoff's Voltage Law (KVL): The sum of all voltage drops around any closed loop in a circuit must be zero.



$$-V_S + V_{R_1} + V_{R_2} = 0$$

or equivalently

$$V_S = V_{R_1} + V_{R_2}$$

A Hiker's Analogy: Suppose we were hiking over hilly terrain. As we move along the hiking trail, we keep track of the changes in elevation (analogous to voltage rises/drops in our circuit). If we hike along a trail that eventually loops back to the starting point, the sum of all the elevation changes we encounter along the way must be zero.

Note:

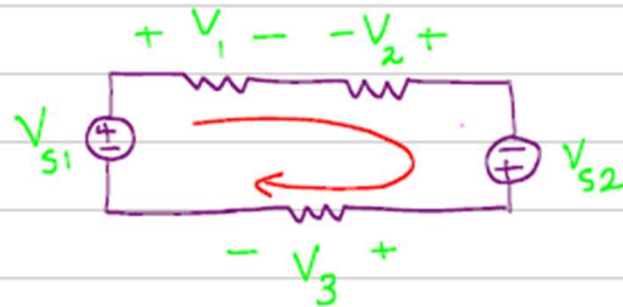
- ✓ it doesn't matter where we start and end on the loop
- ✓ nor does it matter in which direction we traverse the loop.
- ✓ It DOES matter that we measure the voltage the same direction on each element

KVL



Kirchhoff's Voltage Law (KVL): The algebraic sum of all voltages around any loop in the circuit is zero.

$$\sum_{n=1}^N V_n = 0$$



Taking a voltage drop (from + to -) as a positive voltage:

KVL: $V_1 - V_2 - V_{s2} + V_3 - V_{s1} = 0$

Example1



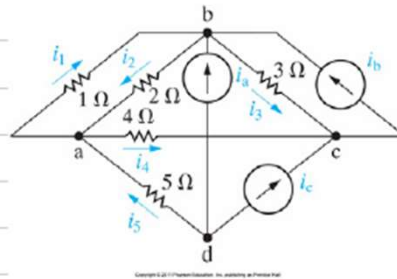
Ex. Write KCL at each node.

$$a: i_1 - i_2 + i_4 - i_5 = 0$$

$$b: -i_1 + i_2 - i_3 + i_5 = 0$$

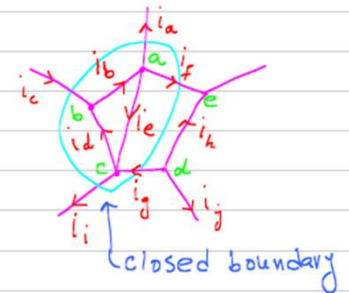
$$c: -i_3 + i_6 - i_4 - i_5 = 0$$

$$d: i_3 + i_4 + i_5 = 0$$



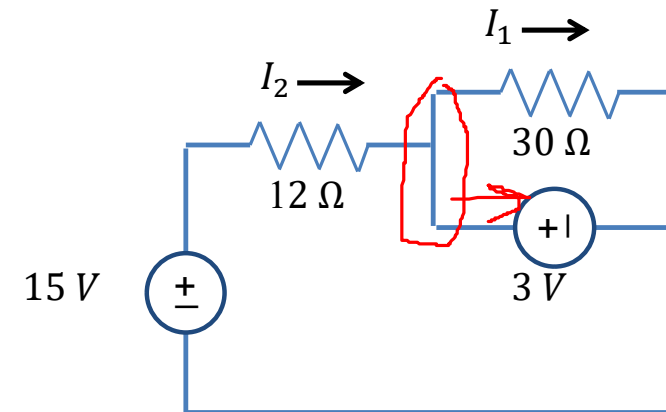
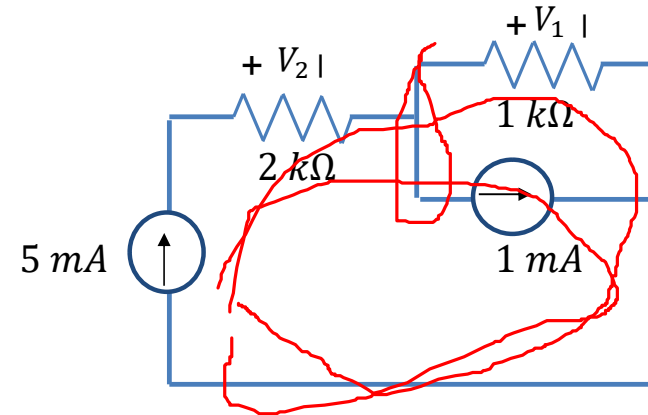
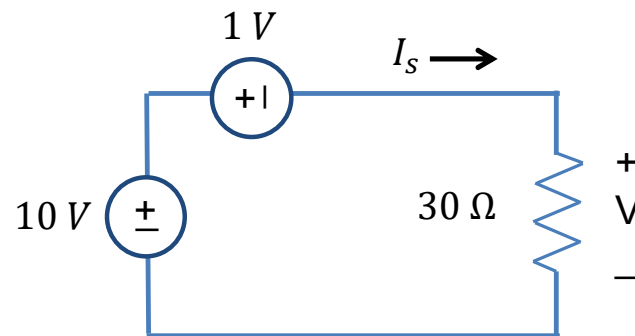
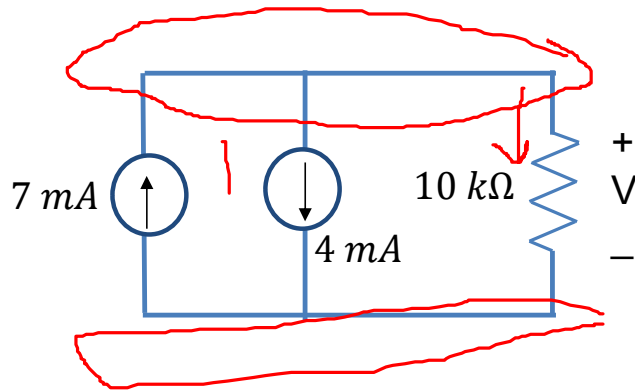
KCL at a closed boundary

$$\begin{cases} a: i_a + i_f - i_b + i_e = 0 \\ b: i_b - i_c - i_d = 0 \\ c: i_d - i_e - i_g + i_i = 0 \end{cases}$$



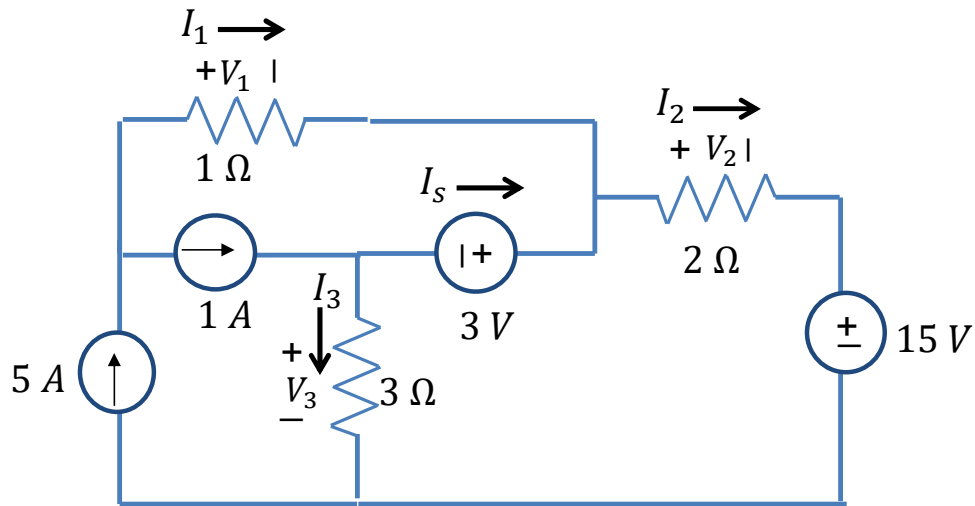
$$\text{Add: } i_a + i_f - i_c - i_g + i_i = 0 \quad (\text{KCL at closed boundary})$$

Apply Kirchhoff's Laws



A Bigger Example

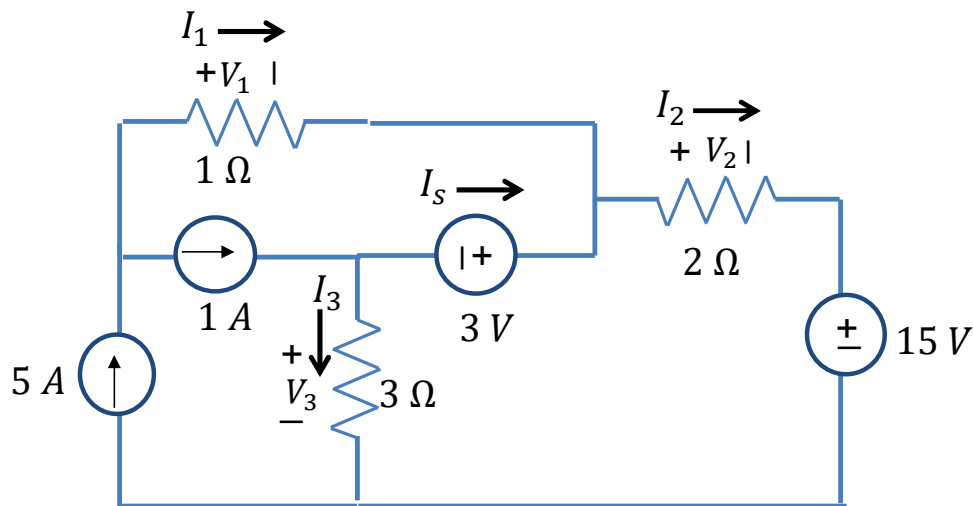
- Write out Ohm's law for each resistor and KCL and KVL everywhere in the circuit that might be useful and see if you can get a set of equations to solve for $V_1, V_2, V_3, I_1, I_2, I_3$



Bonus: what's the voltage across the 5 A current source?

A Bigger Example (2)

- Write out Ohm's law for each resistor and KCL and KVL everywhere in the circuit that might be useful and see if you can get a set of equations to solve for $V_1, V_2, V_3, I_1, I_2, I_3$



Solution:

Ohm's law

$$V_1 = I_1$$

$$V_2 = 2I_2$$

$$V_3 = 3I_3$$

KCL

$$5 - 1 = I_1$$

$$1 = I_s + I_3$$

$$I_1 + I_s = I_2$$

KVL

$$V_3 + 3 - V_2 - 15 = 0$$

$$I_1 = 4 \text{ A}; V_1 = 4 \text{ V}$$

$$I_s = 1 - I_3$$

$$4 + 1 - I_3 = I_2$$

$$3I_3 + 3 - 2I_2 - 15 = 0$$

$$3I_3 + 3 - 2(5 - I_3) - 15 = 0$$

$$I_3 = 4.4 \text{ A}; V_3 = 13.2 \text{ V}$$

$$I_2 = 0.6 \text{ A}; V_2 = 1.2 \text{ V}$$

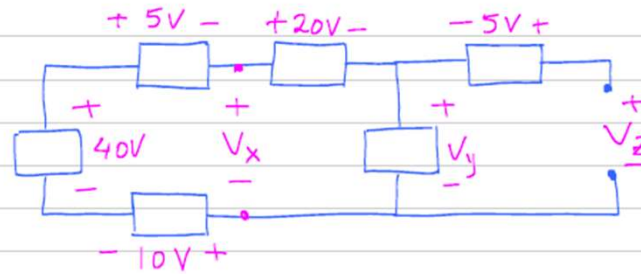
$$I_s = -3.4 \text{ A}$$

Bonus: what's the voltage across the 5 A current source?

Example 3



Ex. Find V_x , V_y and V_z .



$$V_x = -5 + 40 - 10 = 25V$$

$$V_y = -20 - 5 + 40 - 10 = 5V$$

$$V_z = 5 + V_y = 5 + 5 = 10$$

$$\text{or: } V_z = -20 - 5 + 40 - 10 = 10$$

Example 4



Ex. Use the basic laws to find i and check your answer using power balance.

KVL:

$$V_{ab} + V_{bc} - 120 = 0$$

Ohm's Law:

$$10i + 50i_1 - 120 = 0 \quad (1)$$

KCL at b :

$$-i + i_1 - 6 = 0 \quad (2)$$

From (1) & (2):

$$\underline{i = -3A}, \quad i_1 = 3A$$

Check:

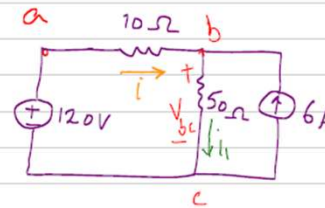
$$P_{120V} = -120(-3A) = 360W$$

$$P_{10\Omega} = Ri^2 = (10)(-3)^2 = 90W$$

$$P_{50\Omega} = Ri_1^2 = (50)(3)^2 = 450W$$

$$P_{6A} = -(6)(V_{bc}) = -6(50 \times 3) = -900W$$

$$360 + 90 + 450 - 900 = 0 \quad \checkmark$$



Example of a Dependent Source

Kirchhoff's Current Law

$$9 + 0.1I_x - I_x = 0$$

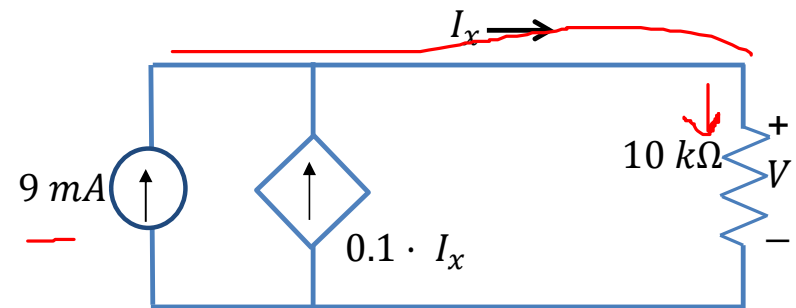
$$I_x = \frac{9}{0.9} = 10 \text{ mA}$$

$$V = 10 \text{ mA} \cdot 10 \text{ k}\Omega = 100 \text{ V}$$

Power analysis

- Independent voltage source producing $(9 \text{ mA})(100 \text{ V}) = 900 \text{ mW}$
- Dependent voltage source producing $(1 \text{ mA})(100 \text{ V}) = 100 \text{ mW}$
- Resistor consuming $(10 \text{ mA})(100 \text{ V}) = 1000 \text{ mW} = 900 \text{ mW} + 100 \text{ mW}$

This is conservation of energy.



Example2 of a Dependent Source



Ex. Find V_o and I_o .



$$V_1 - V_o = 0$$

$$2i_o - 8i_1 = 0 \quad (1)$$

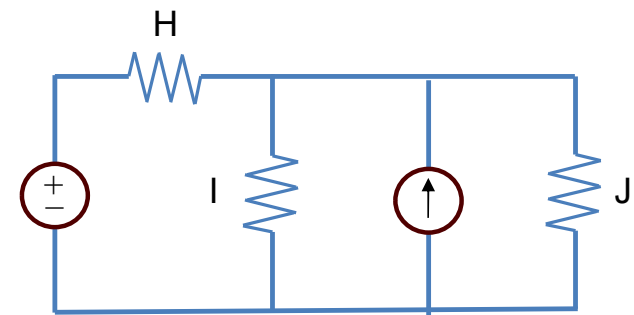
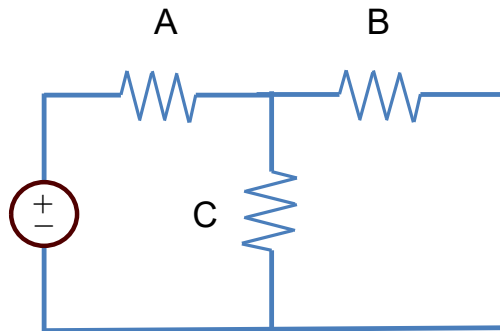
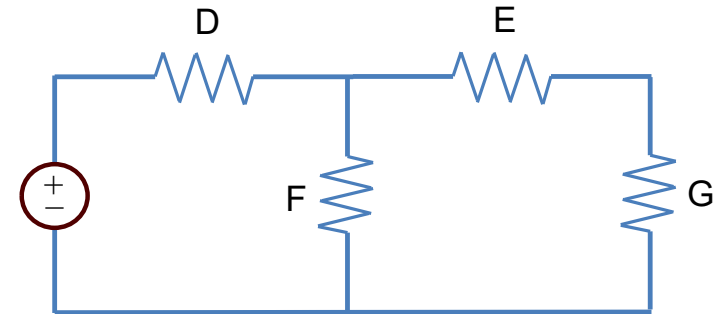
$$\text{KCL: } -6 + i_o + \frac{i_o}{4} + i_1 = 0 \quad (2)$$

$$(1), (2) \Rightarrow i_o = \boxed{4A}, i_1 = 1A$$

$$V_o = 8i_1 = \boxed{8V}$$

Example Parallel or Series?

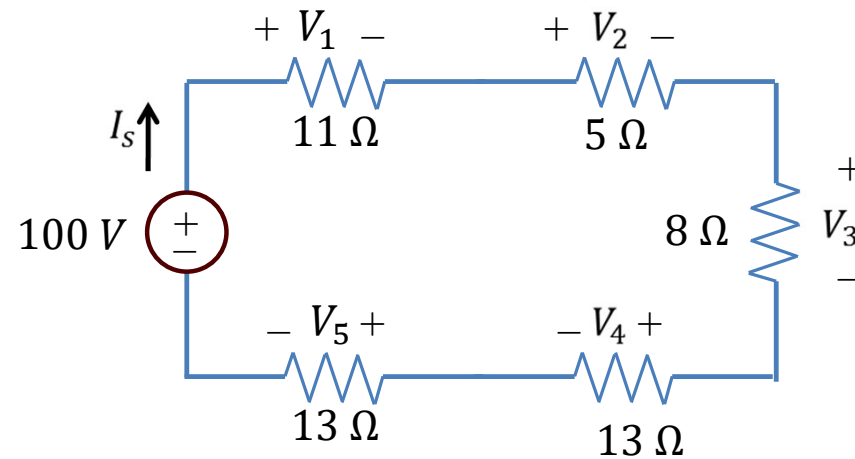
- Elements in **series** share one junction, with nothing else connected to the same junction, and they have the **same current** (KCL)
- Elements in **parallel** share two junctions, regardless of what else is connected, and they have the **same voltage** (KVL)
- Which resistors in these circuits are in series and parallel?



Example with Resistors in Series



- Solve for I_s and the voltage across each resistor



Resistors in Series

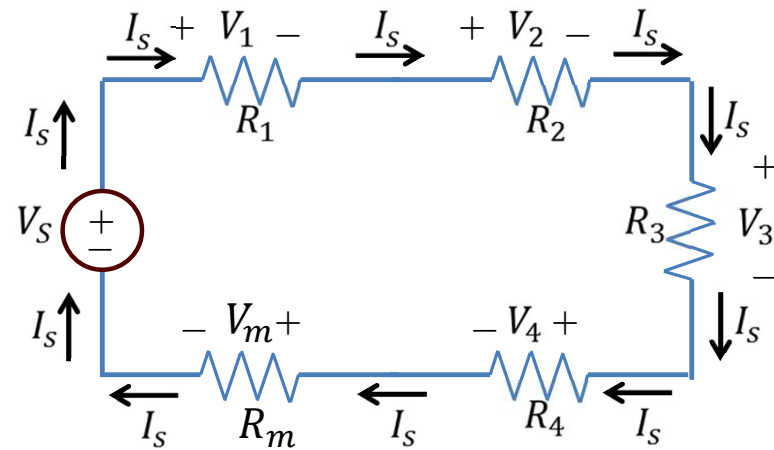
- Consider a circuit with two or more resistors connected in series.
- The current through each resistor must be the same.
- The current through each resistor must be the same.
- By virtue of Ohm's law, we then get:

$$V_1 = I_s R_1$$

$$V_2 = I_s R_2$$

...

$$V_m = I_s R_m.$$



Using Kirchoff's Voltage Law (KVL) around the loop:

$$\begin{aligned} V_s &= V_1 + V_2 + \dots + V_m \\ &= I_s R_1 + I_s R_2 + \dots + I_s R_m \\ &= I_s (R_1 + R_2 + \dots + R_m) \\ &= I_s R_{eq} \end{aligned}$$

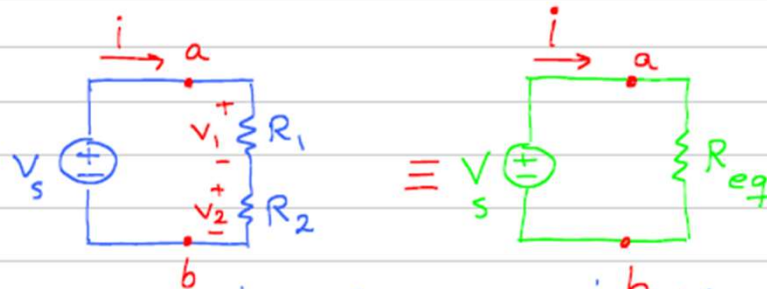
$$R_{eq} = R_1 + R_2 + \dots + R_m$$

Ch2: Resistive Circuits



Equivalent circuits: Two circuits are said to be equivalent if they have the same v-i characteristics at the specified pair of terminals.

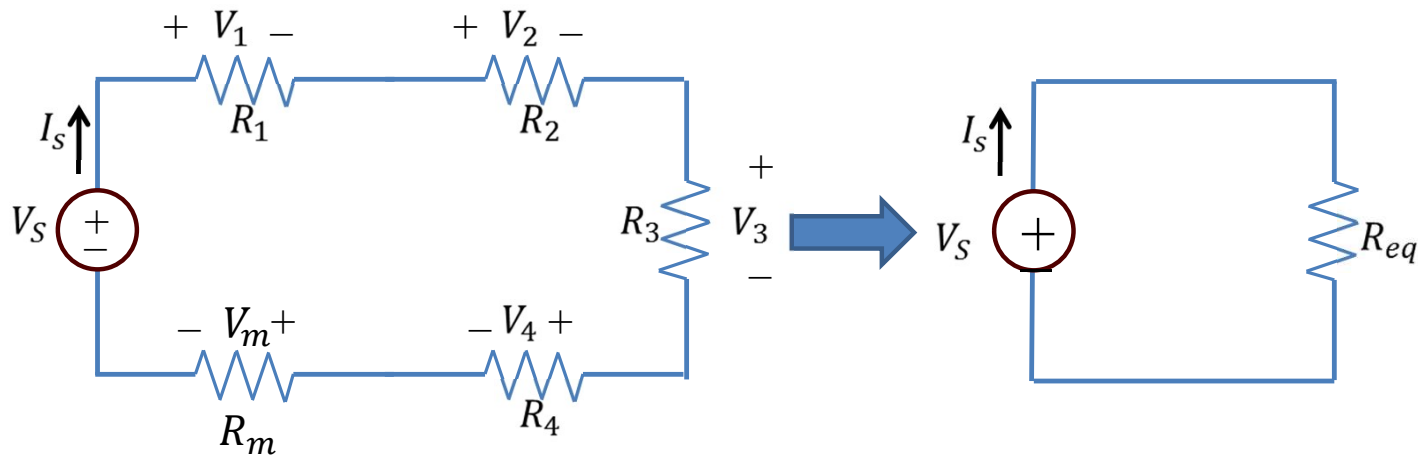
Equivalent of series resistors:



$$\left\{ \begin{array}{l} -V_s + R_1 i + R_2 i = 0 \Rightarrow i = \frac{V_s}{R_1 + R_2} \\ i = \frac{V_s}{R_{eq}} \end{array} \right.$$

$$\Rightarrow \frac{V_s}{R_1 + R_2} = \frac{V_s}{R_{eq}} \Rightarrow \boxed{R_{eq} = R_1 + R_2}$$

Resistors in Series



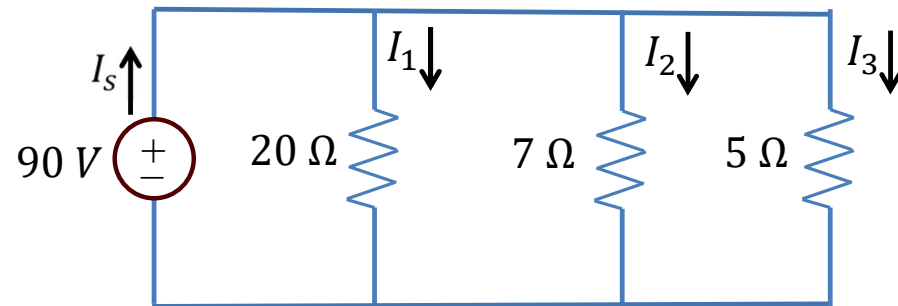
- For the purposes of finding the current flowing through the circuit, the series connection of the multiple resistors can be replaced by a single equivalent resistor whose resistance is the sum.

$$R_{eq} = R_1 + R_2 + \cdots + R_m$$

Example with Resistors in Parallel



- Solve for I_s and the current through each resistor



Resistors in Parallel

- Consider a circuit with two or more resistors connected in parallel.
- Now the current in each branch is different but the voltage across each resistor is the same.

$$V_s = I_1 R_1 = I_2 R_2 = \dots = I_m R_m$$

- By virtue of Ohm's law, we then get:

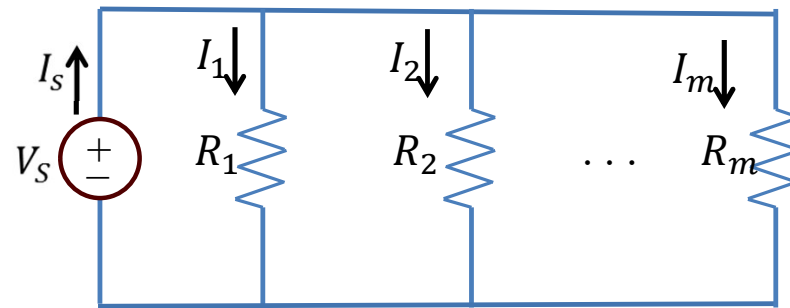
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_m}$$

or equivalently

$$R_{eq} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_m}}$$

or, in terms of conductance:

$$G_{eq} = G_1 + G_2 + \dots + G_N = \sum_{i=1}^N G_i$$

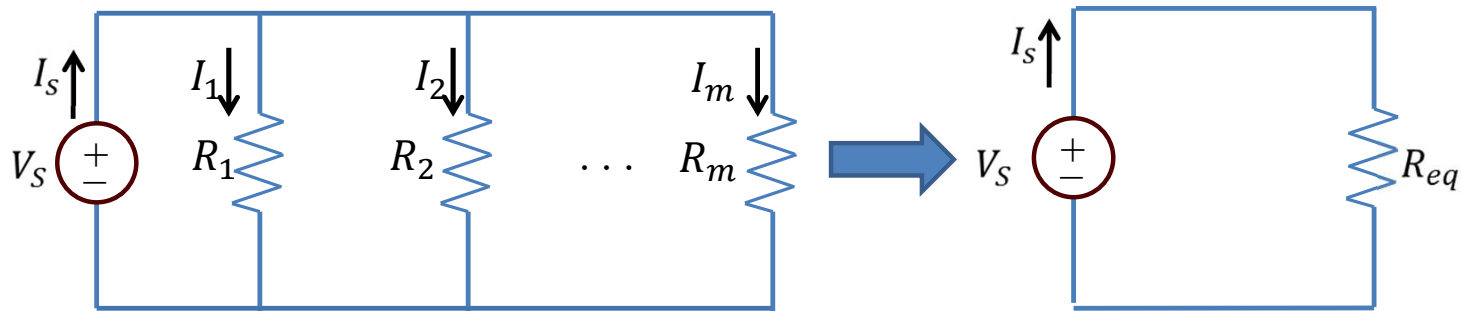


Using Kirchoff's Current Law (KCL):

$$\begin{aligned} I_s &= I_1 + I_2 + \dots + I_m \\ &= \frac{V_s}{R_1} + \frac{V_s}{R_2} + \dots + \frac{V_s}{R_m} \\ &= V_s \cdot \left(\frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_m} \right) \\ &= \frac{V_s}{R_{eq}} \end{aligned}$$

Equivalent Resistances

- Resistors in Parallel



- For the purposes of finding the current flowing through the source, the parallel connection of resistors can be replaced with a single equivalent resistor such that:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_m} \quad \Rightarrow \quad R_{eq} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_m}}$$



Equivalent Resistances

- Most often, we use these concepts with just two resistors at a time. In that case, the formulas simplify to:

Series combination

$$R_{eq} = R_1 + R_2$$

Parallel combination

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

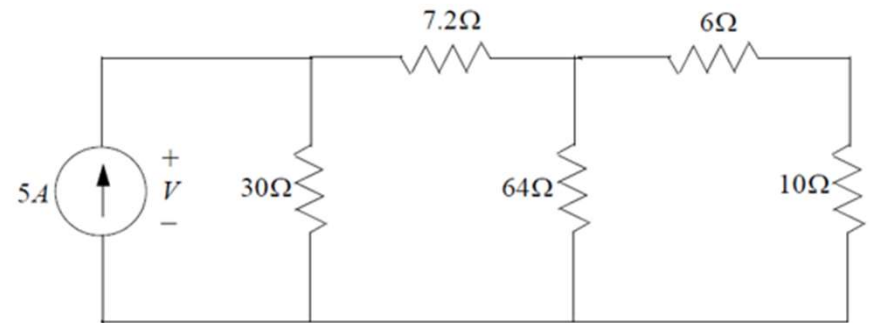
Note that:

Series equivalent resistances are always larger than each individual component, and
Parallel equivalent resistances are always smaller than each individual component,

Example



- For the circuit shown:
 - (a) Find the voltage V ,
 - (b) Find the power delivered to the circuit by the current source,
 - (c) Find the power dissipated in the 10Ω resistor.



Voltage Dividers

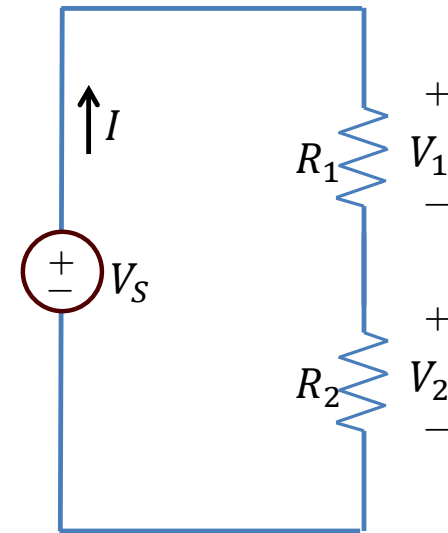
- If a voltage is applied across a series resistance combination, the voltage is divided (unequally) across the resistors.

- $$V_s = V_1 + V_2 = IR_1 + IR_2 \quad I = \frac{V_s}{R_1 + R_2} \quad \Rightarrow$$

$$V_1 = IR_1 = V_s \frac{R_1}{R_1 + R_2}$$

$$V_2 = IR_2 = V_s \frac{R_2}{R_1 + R_2}$$

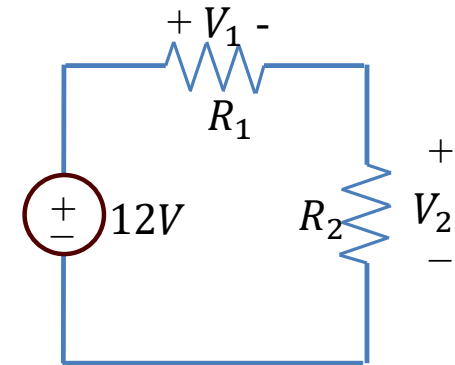
These ratios are known as voltage dividers



- The voltage is split proportionally according to the resistance values. Higher voltage goes to the larger resistor.
- Note:** This kind of configuration occurs so frequently that it is worthwhile committing this result to memory.

Example of Voltage Dividers

- For the voltage divider circuit shown:
 - If $R_1 = 1k\Omega$ and $R_2 = 2k\Omega$, determine the voltages V_1 and V_2 . Also determine the power generated by the source.
 - Suppose we want $V_2 = 9V$ and that the source is capable of producing no more than 250mW of power. What are the smallest resistors that can be used to achieve these specifications?



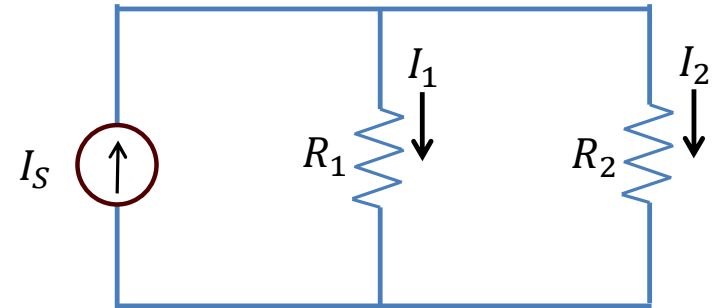
Current Dividers

- If a current is applied to a parallel resistance combination, the current is divided (unequally) through the resistors.
- $I_1 R_1 = I_2 R_2 = I_s R_{eq} = I_s \frac{R_1 R_2}{R_1 + R_2}$

$$I_1 = I_s \frac{R_2}{R_1 + R_2}$$

$$I_2 = I_s \frac{R_1}{R_1 + R_2}$$

These ratios are known as current dividers

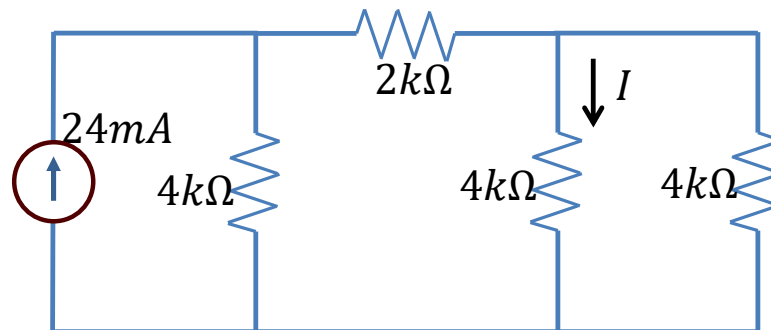


- The current is split proportionally according to the resistance values. Higher current goes to the **smaller** resistor. Again, commit this result to memory.
- Beware:** The current dividers are the opposite of the voltage dividers. This will be a source of great confusion to some.

Example of Current Dividers

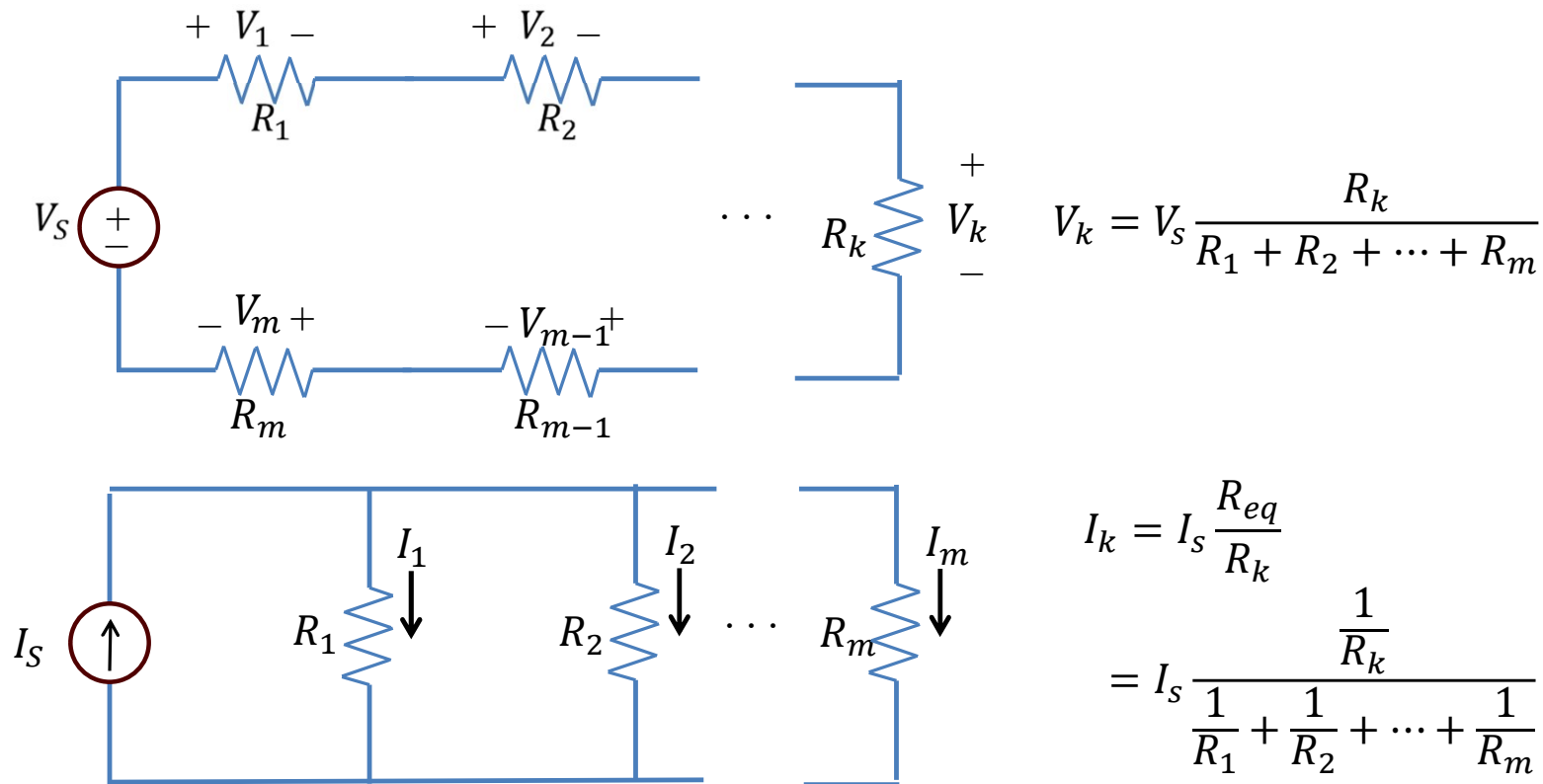


- Find the current I in the circuit shown.



Voltage and Current Dividers

- The concepts of voltage and current division can be applied to more than two resistors.



Review: Main Things You Should Know Now



- Basic circuit elements
 - Independent current source – constant current going through it
 - Independent voltage source – constant voltage across it
 - Resistor – linear relationship between voltage and current through Ohm's law $V=IR$
 - Definitions of current and voltage
 - Kirchhoff's laws
 - Kirchhoff's current law – sum of currents into node must be zero
 - Kirchhoff's voltage law – sum of voltages around any closed loop must be zero
 - Watch your signs carefully in solving circuits
-

Next Lecture



- Practical sources
- Circuit Analysis